FracVal Aggregation Methodology

Sunday, November 8, 2026 8:56 PM

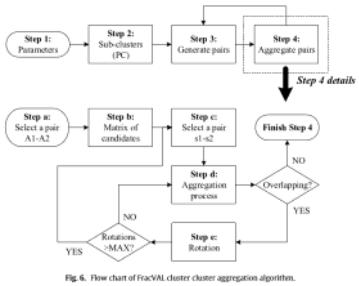


Fig. 6. Flow chart of FracVAL cluster cluster aggregation algorithm.

Step 1:

↳ Pick D_f, k_f, this will be presered throughout the aggregation process

↳ Specify the primary particle size distribution, pick all the N primary particles

Step 2:

↳ use particle-cluster aggregation method to build subclusters that all have similar N_{sub} where $\Sigma N_{sub} = N$ and each subcluster maintains D_f, k_f

Step 3:

↳ organize the subclusters into pairs that are able to aggregate.

① we can use the scaling law to find the required R_g of the new aggregate. $R_g = r_{p,geo}\left(\frac{n_i}{k_f}\right)^{1/D_f}$ ← Prescribed

② we can use conservation of momentum to find the required distance between the COMs of each subcluster to meet R_g

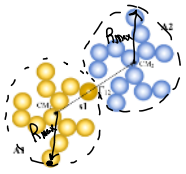
$r_{1,2}$ = distance between the centers of mass of two subclusters forming a new aggregate

$$r_{1,2} = \frac{1}{\sqrt{m_1 m_2}}\left[m_{tot}^2 R_g^2 - m_{tot}(m_1 R_{g_1}^2 + m_2 R_{g_2}^2)\right]^{1/2}$$

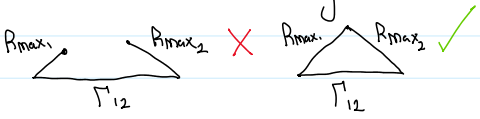
③ check if each pair even has the potential to aggregate by using $R_{max,1} + R_{max,2} \geq r_{1,2}$ * we do not care if they overlap here, just could they touch?

R_{max} is the furthest distance of a primary particle to its subcluster COM

* this is using the triangle inequality



⇒ Can $R_{max,1,2}$ and r_{12} form a triangle?



* Note this is checking based on geometry to see if r_{12} can be maintained only. It does not check for overlaps/actual physical ability of aggregation

Step 4: the heart of the algorithm ♡

↳ in this step we are forming a larger aggregate from our pairs of subclusters

↳ the two subclusters must be combined such that r_{12} is maintained. There are multiple solutions, but we just need to pick one.

↳ There are six substeps:

① Pick a pair of subclusters from step 3

i.e. can point touch and meet r_{12} req.

↳ There are six substeps:

① Pick a pair of subclusters from step 3

② So now we need to determine which primary particles can collide. We go through every combination of primary particles between the clusters. The primary particles are close enough to be point touching while maintaining Γ if:

$D_{i1+} + D_{j2+} \geq \Gamma_{12}$ (i.e. can point touch and meet Γ_{12} req.)

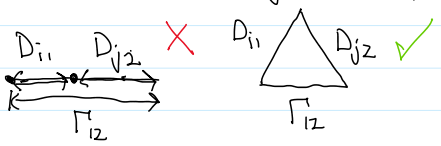
Where $D_{i1} = d_{i1} + r_{pi}$

↑ distance from cluster COM to PP COM; ↑ radius of PP

* Basically triangle inequality again, except now we are also checking the lower limit too to ensure that the particles genuinely overlap and one does not just completely surround the other

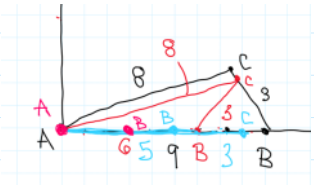
▷ Lower limits:

case 1) $|D_{i1} - D_{j2}| \leq \Gamma_{12}$

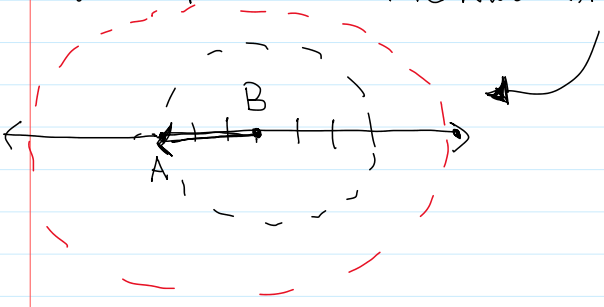


Standard case, Particles can point touch

c (=Γ_{12})	A coords	B coords	C coords	a (fixed)	b (fixed)	check: d(C,A)	check: d(C,B)
9	(0, 0)	(9, 0)	(7.56, 2.63)	8.0	3.0	8.00	3.00
6	(0, 0)	(6, 0)	(7.58, 2.55)	8.0	3.0	8.00	3.00
5	(0, 0)	(5, 0)	(8.00, 0.00)	8.0	3.0	8.00	3.00
3	(0, 0)	(3, 0)	NO SOLUTION	8.0	3.0	---	---



What failure of the lower limit means - complete engulfment of one sphere.



However, while case 1 fails, there is still a potential point contact between the engulfing sphere's inner edge, D_{i1-} and the engulfed cluster's outer reach, D_{j2+}. This and it's analogous counterpart become cases 2+3

Cases 2 + 3:

$|D_{i1+} - D_{j2+}| > \Gamma_{12}$ and an upper limit of $D_{i1-} \leq \Gamma_{12} + D_{j2+}$ and analogously $D_{j2-} \leq \Gamma_{12} + D_{i1+}$

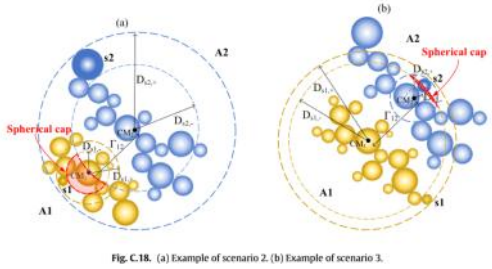
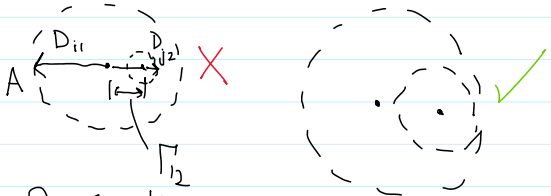


Fig. C.18. (a) Example of scenario 2. (b) Example of scenario 3.

* Cases 2 and 3 are edge cases fractal handles, but Case 1 is most common and will be used to explain the remainder of the steps



So if case 1, 2, or 3 are satisfied, the primary particles point touch and we mark a 1 for a_{ij} in our matrix. If not, we mark a 0 meaning no point touch is possible.

③ Okay, so ② was just determining given D_{i1+}, D_{j2+}, and Γ_{12} is an intersection even possible for our spheres? So we loop over the elements of our matrix A. We pick the first pair. We then start setting up the 3D Geometry. it must satisfy 3 requirements:

So we loop over the elements of our matrix A. We pick the first pair. We then start setting up the 3D Geometry. it must satisfy 3 requirements:

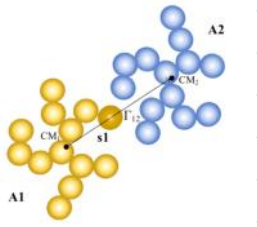
1) the distance between CM1 and CM2 must be Γ_{12}
2) the two primary particles are in point touch
3) no overlapping between any primary particles in A1+A2

this is done in 3 substages:

▶ Substage 1:

↳ we treat CM1 as fixed and make it our new origin.

↳ We place CM2, Γ_2 away from CM1 by placing it along the direction defined by CM1 and the center of s1



▶ Substage 2:

↳ We next place spheres one w/ radius D_{s1+} at CM1 and D_{s2+} at CM2. the intersection of these two spheres will give the location of the center of s1. Note that the intersection of the two spheres creates a ring of possible options. We parameterize the circle and pick an angle randomly which gives a point on the circle. From here we rotate A1 to place S1.

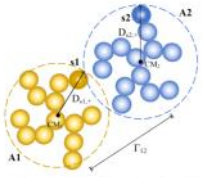


Fig. 4. Location of monomer s1 given by the intersection of spheres given by Eqs. (11a) and (11b).

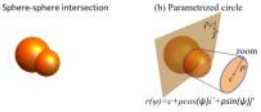


Fig. 5. Sphere–sphere intersection and circle parametrization.

$$D_{s1+}^2 = (X - X_{CM_1})^2 + (Y - Y_{CM_1})^2 + (Z - Z_{CM_1})^2$$
$$D_{s2+}^2 = (X - X_{CM_2})^2 + (Y - Y_{CM_2})^2 + (Z - Z_{CM_2})^2$$

$$\vec{r}[\psi] = \vec{c} + \rho\cos[\psi]\hat{i} + \rho\sin[\psi]\hat{j}, \quad \psi \in [0, 2\pi]$$

just need to pick a ψ

see Appendix D for derivation

▶ Substage 3

↳ We now need to place the center of s2 to lock in the geometry. We placed the center of s1 to ensure the possibility of point contact, so now we actually do the math to get the center of s2 into the correct spot. We need s1 and s2 to be in point contact and the center of s2 to be d_{s2} distance from CM 2. So we place a sphere at the center of s1 with radius $(r_{s1}+r_{s2})$ and a sphere of radius d_{s2}. This will form a ring of solutions and so we again parameterize the circle and pick an angle like before.

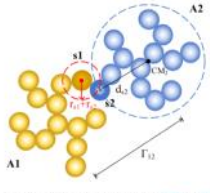


Fig. 7. Intersection of spheres given by Eqs. (13a) and (13b).

$$(r_{s1} + r_{s2})^2 = (X - X_{s1})^2 + (Y - Y_{s1})^2 + (Z - Z_{s1})^2$$
$$d_{s2}^2 = (X - X_{CM_2})^2 + (Y - Y_{CM_2})^2 + (Z - Z_{CM_2})^2$$

(4) Now we check if any of the primary particles overlap. The paper does not explicitly state how but they likely compare the distance between the centers of every PP pair and if it is greater than the sum of their radii then there is no overlap. If there is overlap then pick another point on the

not explicitly state how but they likely compare the distance between the centers of every PP pair and if it is greater than the sum of their radii then there is no overlap. If there is overlap, they pick another point on the ring from (3) substep 3 and recheck. If there is still overlap after N iterations it will go back and chose the next pair and try again.